Listing 2.3 An optimization of the bubbleUp method

```
function bubbleUp(pairs,
index=|pairs|-1)                                        #1
  (parentIndex, current) ← (index,
pairs[index])                                  #2
  while parentIndex > 0
do                                                      #3
    currentIndex ← parentIndex
    parentIndex ←
getParentIndex(parentIndex)                                      #4
    (element, priority) ← pairs[currentIndex]
    if pairs[parentIndex].priority < priority then
         #5
      pairs[currentIndex] ←
pairs[parentIndex]                                  #6
    else
                #7
      break
  pairs[parentIndex] ← current
                                               #8
```

#1 We explicitly pass the array with all pairs and the index (by default the last element) as arguments

#2 Start from the element at the index passed as argument (by default the last element of A)

#3 Check if the current element is already at the root of the heap: if so, we are finished

#4 Compute the index of the parent of current element in the heap. The formula can vary with the type of implementation. For a heap with branching factor `D` and array's indices starting at zero, it's `(parentIndex-1) / D`